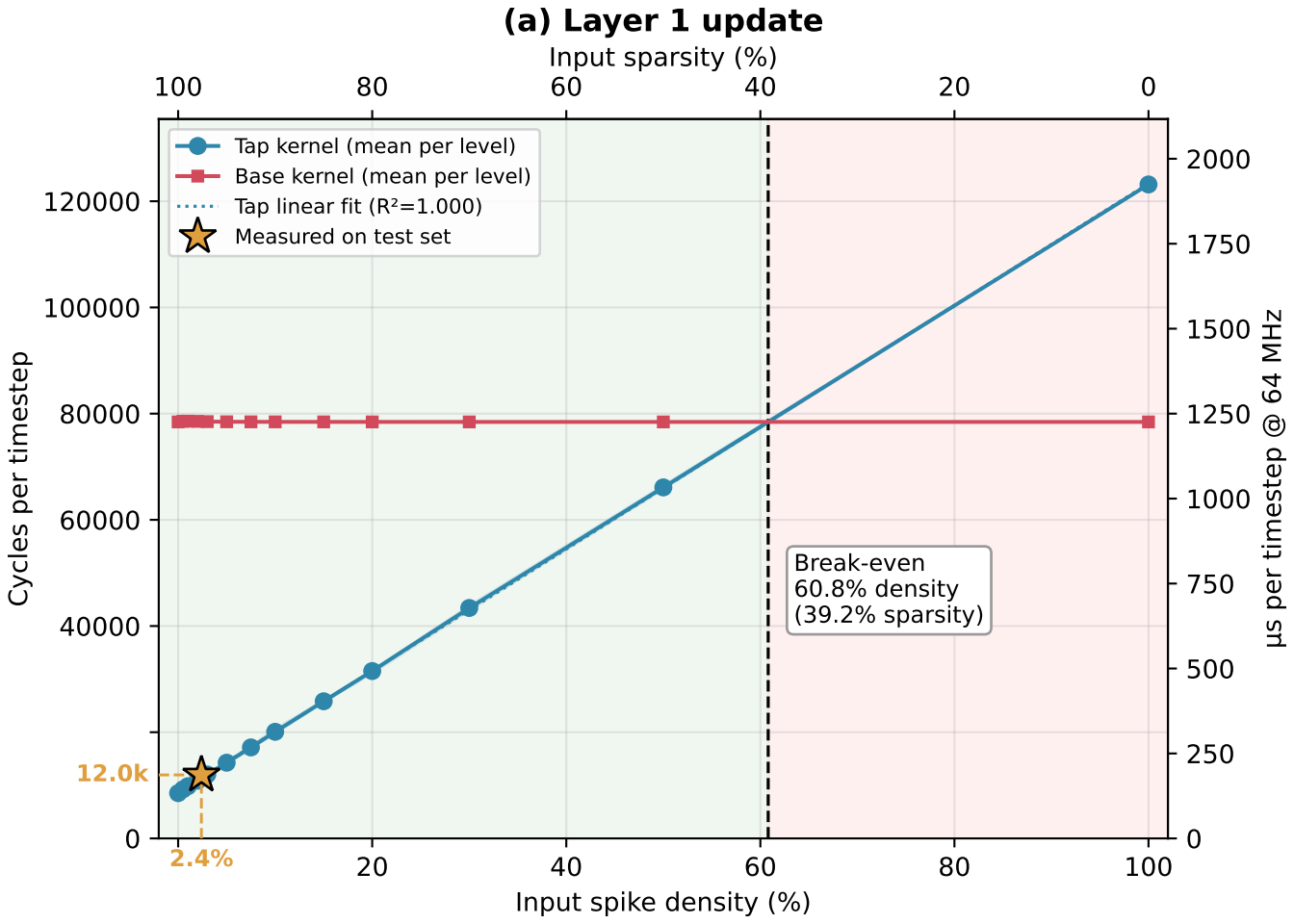
(a) Layer 1 update
Input sparsity (%)
100
80
60
40
20
0
Tap kernel (mean per level)
Base kernel (mean per level)
Tap linear fit (R²=1.000)
Measured on test set
120000
100000
80000
60000
40000
12.0k
0
Cycles per timestep
2000
1750
1500
1250
1000
750
500
250
0
µs per timestep @ 64 MHz
Break-even
60.8% density
(39.2% sparsity)
2.4%
20
40
60
80
100
Input spike density (%)